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| | **ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)** | | **Assignment** |
| Branch | **Computer Science and Engineering, Computer Science and Information Technology** | Programme | **B.Tech** |
| Course Name | **Science and Technology in Ancient India** | Semester | **VIII** |
| Course Code | **IKS4003** | Academic Year | **2026/Even** |
| **Assignment-2** | **Topic- Developments in Science and Technology in Medieval India** | **GP-1** | |

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| Learning Level (LL) | **L1**: Remembering | **L3:** Applying | **L5**: Evaluating |
| | **L2**: Understanding | **L4**: Analyzing | **L6**: Creating |

| Q's | Questions | COs | LL |
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| **1** | Provide one reason why the encephalization factor can or cannot be viewed as a definitive trait of superiority. | **CO1** | **L9** |
| **2** | How does the dharma tradition integrate physical and metaphysical concepts within the panchakosa-based evolution system? | **CO1** | **L10** |
| **3** | How many seasons are generally mentioned in the Vedic Samhitas? Which planets were known by name in the Vedic period? | **CO1** | **L10** |
| **4** | Briefly explain the role of "nadi" and "prana" in the operational aspects of reincarnation. | **CO1** | **L10** |
| **5** | What is the earliest evidence of copper use in India? | **CO3** | **L6** |
| **6** | How did Harappans use copper and bronze? | **CO3** | **L6** |
| **7** | Define *Purushartha* and explain how it serves as a "live demonstration" as described by an acharya. | **CO2** | **L9** |
| **8** | How does the practice of yoga allow an individual to experientially observe the chakra system? | **CO2** | **L9** |
| **9.** | How was gold obtained and used in ancient India? What is the significance of silver production in ancient India? | **CO3** | **L6** |
| **10** | Explain the metaphor of logic as the "cement" between the two stones of science and spirituality. | **CO3** | **L7** |
| **11** | When did iron technology emerge in India? What is wootz steel and why was it important? | **CO3** | **L7** |
| **12** | Briefly explain the role of technology in shaping the concept of a "Global Village". | **CO2** | **L10** |
| **13** | What is the significance of the Delhi Iron Pillar? | **CO3** | **L8** |

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| 14 | Why is India significant in zinc production? | CO3 | L8 |
| 15 | Distinguish between *Para-vidya* and *Apara-vidya* as presented in the Upanishads. | CO2 | L10 |
| 16 | What is bidri ware and how does zinc feature in it? Who controlled metallurgical practices in ancient India? | CO3 | L8 |
| 17 | Identify the similarity between Sankara's "existence of being" and Hegel's "pure being". | CO2 | L9 |
| 18 | Briefly summarize the notion of "absolute spirit" according to Rene Descartes and Spinoza. | CO2 | L9 |
| 19 | What does the Arthasastra reveal about metallurgy? | CO3 | L6 |
| 20 | How did Varahamihira contribute to metallurgy?Explain how the "Absolute" is conceptualized in the *Nasadiya-Sukta* of the Rigveda. | CO3 | L6 |
| 21 | List the five great teachings (mahavakyas) from the Upanishads regarding the relationship between the Self and God. | CO2 | L9 |
| 22 | Based on the Kenopanishad and Kathopanishad, what happens to a person who fails to attain Self-realization before death? | CO2 | L10 |
| 23 | Contrast the "Universal" path of local spirit philosophies (Way X) with the path prescribed by a Guru (Way Y). | CO2 | L10 |
| 24 | Discuss the challenge of accepting a Guru's path (Way Y) in a society characterized by multiple faiths and diverse opinions. | CO2 | L10 |

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| **Assignment 2** | **Topic: Developments in Science and Technology in Medieval India** | **Date of Assignment1: 16.03.2026** | **Date of Submission: 23.03.2026** |

**Note:**

1. assignment carries weightage of **20 marks out of 100**
2. Course outcome CO1 to CO3 was covered.

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| Course Outcomes | CO1 | To provide a comprehensive overview of the historical development of scientific thought and technological practices in ancient India, from the Vedic period to the medieval era. |
| | CO2 | To analyse ancient Indian contributions to fundamental sciences, Mathematics (*Sulba − Sutras*), Astronomy (*Jyotisha*), and Physics. |
| | CO3 | To evaluate technological advancements in fields such as Metallurgy, Civil Engineering, and Architecture (Vastu Shastra). |
| | CO4 | To examine the evolution of medicinal systems, specifically the theories and surgical practices found in the foundational texts of Ayurveda |
| | CO5 | To foster a critical understanding of how ancient Indian scientific knowledge was documented in Sanskrit texts and its subsequent influence on global scientific thought through translation and trade. |
| | CO6 | To investigate the Vedic and Post-Vedic understanding of heat transfer, latent heat, and the role of Agni in the phase transitions of matter (solid to liquid to gas). |